

17.3 Separating Hyperplanes

Sections 01 and 02 looked at a convex set from the inside: which combinations of its points it contains, and how few points a combination needs. This section looks at it from the outside. A point that is not in a closed convex set can be cut off from it by a hyperplane, a set can be touched at its edge by a hyperplane that leaves the whole set on one side, and two disjoint convex sets can be pulled apart by one. All three facts come from a single construction, the point of the set nearest to a given point, and that construction is Chapter 10's orthogonal projection with the subspace replaced by a convex set.

Throughout, V is a finite-dimensional real inner product space, with inner product $\langle \cdot, \cdot \rangle$ (Definition 10.1.1), induced norm $\|\cdot\|$ (Definition 10.1.3), and $n = \dim V$. A sequence **converges**, $\mathbf{x}_k \rightarrow \mathbf{x}$, when $\|\mathbf{x}_k - \mathbf{x}\| \rightarrow 0$; **closed** means closed in the sense of Definition 17.1.6, so that a closed set contains the limit of every convergent sequence of its points. A set is **compact** if every sequence in it has a subsequence converging to a point of the set; this is the word of Corollary 15.2.5, which says that every closed bounded subset of V is compact. The field is $\mathbb{R}$ because the whole section is about inequalities.

17.3.1 The nearest point of a closed convex set

Chapter 10 answered a question about subspaces: given $\mathbf{v}$ and a finite-dimensional subspace U , which vector of U is closest to $\mathbf{v}$? Theorem 10.3.5 said there is exactly one, the orthogonal projection $P_U \mathbf{v}$, and that $\mathbf{v} - P_U \mathbf{v}$ is perpendicular to U . Now replace U by a set with less structure, such as a disc, a box or a half-plane, and ask the same question. Two small experiments in $\mathbb{R}^2$ show which hypotheses the answer needs.

- The **open** disc $\{\mathbf{y} : \|\mathbf{y}\| < 1\}$ and the point $\mathbf{x} = (2, 0)$. The points $(1 - 1/k, 0)$ of the disc come closer and closer to $\mathbf{x}$, but the distance 1 is never reached, since the only candidate $(1, 0)$ is missing. There is no nearest point. The set is convex but not closed.
- The **circle** $\{\mathbf{y} : \|\mathbf{y}\| = 1\}$ and the point $\mathbf{x} = \mathbf{0}$. Every point of the circle is at distance 1, so every point is a nearest point. The set is closed but not convex.

So we ask for both properties, and then everything works.

Theorem 17.3.1 — Nearest Point Theorem

Let C be a **non-empty, closed, convex** subset of V , and let $\mathbf{x} \in V$.

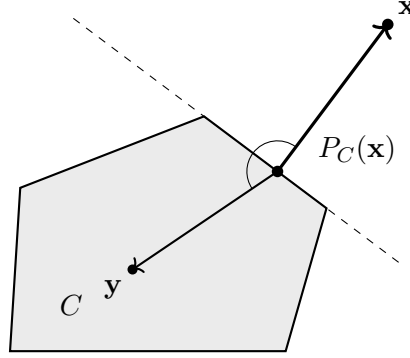
- There is **exactly one** point $\mathbf{p} \in C$ with $\|\mathbf{x} - \mathbf{p}\| \leq \|\mathbf{x} - \mathbf{y}\|$ for every $\mathbf{y} \in C$.
- A point $\mathbf{p} \in C$ is this nearest point **if and only if**

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle \leq 0 \quad \text{for every } \mathbf{y} \in C. \quad (17.3.1)$$

The nearest point is written $P_C(\mathbf{x})$, and the map $P_C: V \rightarrow C$ is the **projection onto C** . The number $d(\mathbf{x}, C) = \|\mathbf{x} - P_C(\mathbf{x})\|$ is the **distance** from $\mathbf{x}$ to C .

Idea. Three jobs, three tools. ① Existence: a nearest point is a minimum of the continuous function $\mathbf{y} \mapsto \|\mathbf{x} - \mathbf{y}\|$, and a minimum exists on a compact set; C itself may be unbounded, so cut it down to the part of C within a fixed distance of $\mathbf{x}$. ②

The inequality Eq. 17.3.1 is the first-order condition: slide from $\mathbf{p}$ a little way towards another point $\mathbf{y}$ of C , which convexity allows, and ask that the distance not decrease. ③ Read backwards, the inequality says that the angle at $\mathbf{p}$ between $\mathbf{x} - \mathbf{p}$ and $\mathbf{y} - \mathbf{p}$ is at least a right angle, and expanding $\|\mathbf{x} - \mathbf{y}\|^2$ then gives $\|\mathbf{x} - \mathbf{y}\|^2 \geq \|\mathbf{x} - \mathbf{p}\|^2 + \|\mathbf{y} - \mathbf{p}\|^2$. That inequality proves both that $\mathbf{p}$ is nearest and that nothing else is.



the angle at $P_C(\mathbf{x})$ between $\mathbf{x} - P_C(\mathbf{x})$ and $\mathbf{y} - P_C(\mathbf{x})$ is obtuse;
the dashed line through $P_C(\mathbf{x})$ is perpendicular to $\mathbf{x} - P_C(\mathbf{x})$

Proof. Existence. Since $C \neq \emptyset$, pick $\mathbf{y}_0 \in C$ and put $R = \|\mathbf{x} - \mathbf{y}_0\|$. Let

$$K = \{\mathbf{y} \in C : \|\mathbf{x} - \mathbf{y}\| \leq R\}.$$

Then $\mathbf{y}_0 \in K$, and K is bounded, since $\|\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y} - \mathbf{x}\| \leq \|\mathbf{x}\| + R$ for $\mathbf{y} \in K$ by Corollary 10.1.6. It is closed: if $\mathbf{y}_k \in K$ and $\mathbf{y}_k \rightarrow \mathbf{y}$, then $\mathbf{y} \in C$ because C is closed, and

$$\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}_k\| + \|\mathbf{y}_k - \mathbf{y}\| \leq R + \|\mathbf{y}_k - \mathbf{y}\|$$

for every k ; the right-hand side tends to R , and a non-strict inequality survives a limit, so $\|\mathbf{x} - \mathbf{y}\| \leq R$. By Corollary 15.2.5, the form in which Chapter 15 carried **the compactness of closed bounded sets, fact (A3) of Chapter 15's introduction**, over to every finite-dimensional space, K is compact. The function $f(\mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ is continuous, since $|f(\mathbf{y}) - f(\mathbf{y}')| \leq \|\mathbf{y} - \mathbf{y}'\|$ by Lemma 15.1.1. By **the extreme value theorem, fact (A4) of Chapter 15's introduction**, f attains a minimum on K at some $\mathbf{p} \in K$. Every $\mathbf{y} \in C \setminus K$ has $f(\mathbf{y}) > R \geq f(\mathbf{p})$. So $\|\mathbf{x} - \mathbf{p}\| \leq \|\mathbf{x} - \mathbf{y}\|$ for every $\mathbf{y} \in C$.

(b), ($\Rightarrow$). Let $\mathbf{p}$ be a nearest point and let $\mathbf{y} \in C$. For $0 < t \leq 1$, the point $\mathbf{p} + t(\mathbf{y} - \mathbf{p}) = (1 - t)\mathbf{p} + t\mathbf{y}$ lies in C by Definition 17.1.1. Hence, expanding with Theorem 10.1.2 (c) over $\mathbb{R}$,

$$\|\mathbf{x} - \mathbf{p}\|^2 \leq \|(\mathbf{x} - \mathbf{p}) - t(\mathbf{y} - \mathbf{p})\|^2 = \|\mathbf{x} - \mathbf{p}\|^2 - 2t\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle + t^2\|\mathbf{y} - \mathbf{p}\|^2.$$

Cancel $\|\mathbf{x} - \mathbf{p}\|^2$ and divide by $2t > 0$: $\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle \leq \frac{t}{2}\|\mathbf{y} - \mathbf{p}\|^2$. This holds for every $t = 1/k$, and letting $k \rightarrow \infty$ gives Eq. 17.3.1.

(b), ($\Leftarrow$), and uniqueness. Let $\mathbf{p} \in C$ satisfy Eq. 17.3.1, and let $\mathbf{y} \in C$. The same expansion with $t = 1$ gives

$$\begin{aligned} \|\mathbf{x} - \mathbf{y}\|^2 &= \|\mathbf{x} - \mathbf{p}\|^2 - 2\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle + \|\mathbf{y} - \mathbf{p}\|^2 \\ &\geq \|\mathbf{x} - \mathbf{p}\|^2 + \|\mathbf{y} - \mathbf{p}\|^2. \end{aligned} \tag{*}$$

So $\|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x} - \mathbf{p}\|$, and $\mathbf{p}$ is a nearest point. Finally, the nearest point $\mathbf{p}$ found in the existence step satisfies Eq. 17.3.1 by $(\Rightarrow)$, so $(*)$ applies to it; for $\mathbf{y} \in C$ with $\mathbf{y} \neq \mathbf{p}$ it gives $\|\mathbf{x} - \mathbf{y}\|^2 \geq \|\mathbf{x} - \mathbf{p}\|^2 + \|\mathbf{y} - \mathbf{p}\|^2 > \|\mathbf{x} - \mathbf{p}\|^2$, and $\mathbf{y}$ is not a nearest point. This proves the theorem. ■

The inequality $(*)$ is Pythagoras with the equality relaxed to an inequality. That is exactly the relation with Chapter 10, and it deserves to be said precisely.

This theorem generalizes Theorem 10.3.5. Let $C = U$ be a subspace of V . It is convex, and closed by Proposition 17.1.6 (c). For $\mathbf{p} \in U$, the vector $\mathbf{y} - \mathbf{p}$ runs over **all** of U as $\mathbf{y}$ does, and with each $\mathbf{u} \in U$ it also takes the value $-\mathbf{u}$. So Eq. 17.3.1 says $\langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle \leq 0$ and $-\langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle \leq 0$, that is,

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle = 0 \quad \text{for every } \mathbf{u} \in U.$$

For a subspace the variational inequality becomes an equality, the statement $\mathbf{x} - \mathbf{p} \in U^\perp$. Then $\mathbf{x} = \mathbf{p} + (\mathbf{x} - \mathbf{p})$ is the splitting of Theorem 10.3.2 (a), so $P_C(\mathbf{x}) = P_U \mathbf{x}$ by Definition 10.3.2, and $(*)$ becomes the equation of Theorem 10.1.3 used in the proof of Theorem 10.3.5. The projection onto a closed convex set is the orthogonal projection with “perpendicular” weakened to “at an obtuse angle”.

Example. (Three projections)

Find $P_C(\mathbf{x})$ in each case, and check Eq. 17.3.1.

- (a) C is the closed unit ball $\{\mathbf{y} : \|\mathbf{y}\| \leq 1\}$ of V , and $\|\mathbf{x}\| > 1$.
- (b) C is the box $[0, 1]^3 = \{\mathbf{y} \in \mathbb{R}^3 : 0 \leq y_i \leq 1 \text{ for each } i\}$ with the dot product, and $\mathbf{x} = (3, -1, 2)$.
- (c) C is the half-space $\{\mathbf{y} : \langle \mathbf{a}, \mathbf{y} \rangle \leq c\}$, where $\mathbf{a} \neq \mathbf{0}$, and $\langle \mathbf{a}, \mathbf{x} \rangle > c$.

Solution.

- (a) $P_C(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|$. With $\mathbf{p} = \mathbf{x}/\|\mathbf{x}\|$ we have $\|\mathbf{p}\| = 1$ and $\mathbf{x} - \mathbf{p} = (\|\mathbf{x}\| - 1)\mathbf{p}$, so for $\mathbf{y} \in C$

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle = (\|\mathbf{x}\| - 1)(\langle \mathbf{p}, \mathbf{y} \rangle - 1) \leq 0,$$

because $\|\mathbf{x}\| - 1 > 0$ and $\langle \mathbf{p}, \mathbf{y} \rangle \leq \|\mathbf{p}\|\|\mathbf{y}\| \leq 1$ by Theorem 10.1.5. By Theorem 17.3.1 (b), $\mathbf{p}$ is the nearest point, and $d(\mathbf{x}, C) = \|\mathbf{x}\| - 1$.

- (b) Clamp each coordinate into $[0, 1]$: $\mathbf{p} = (1, 0, 1)$. Then $\mathbf{x} - \mathbf{p} = (2, -1, 1)$ and, for $\mathbf{y} \in C$,

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle = 2(y_1 - 1) - y_2 + (y_3 - 1) \leq 0,$$

since $y_1 \leq 1$, $y_2 \geq 0$ and $y_3 \leq 1$. So $P_C(\mathbf{x}) = (1, 0, 1)$ and $d(\mathbf{x}, C) = \sqrt{4 + 1 + 1} = \sqrt{6}$. The vector $\mathbf{a} = \mathbf{x} - \mathbf{p} = (2, -1, 1)$ also cuts $\mathbf{x}$ off: $\langle \mathbf{a}, \mathbf{x} \rangle = 9$ and $\|\mathbf{a}\|^2 = 6$, so $c = \langle \mathbf{a}, \mathbf{x} \rangle - \frac{1}{2}\|\mathbf{a}\|^2 = 6$, and the plane $2y_1 - y_2 + y_3 = 6$ strictly separates C from $\mathbf{x}$, since its left side is at most $2 + 0 + 1 = 3$ on C and equals 9 at $\mathbf{x}$. Theorem 17.3.2 below turns this recipe into a theorem.

- (c) Put $\lambda = (\langle \mathbf{a}, \mathbf{x} \rangle - c)/\|\mathbf{a}\|^2$, which is positive, and $\mathbf{p} = \mathbf{x} - \lambda\mathbf{a}$. Then $\langle \mathbf{a}, \mathbf{p} \rangle = \langle \mathbf{a}, \mathbf{x} \rangle - \lambda\|\mathbf{a}\|^2 = c$, so $\mathbf{p} \in C$, and for $\mathbf{y} \in C$

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle = \lambda(\langle \mathbf{a}, \mathbf{y} \rangle - c) \leq 0.$$

So $P_C(\mathbf{x}) = \mathbf{x} - \lambda \mathbf{a}$: drop the perpendicular to the boundary hyperplane, as Chapter 10 would have done. The distance is $\lambda \|\mathbf{a}\| = (\langle \mathbf{a}, \mathbf{x} \rangle - c) / \|\mathbf{a}\|$.

Two degenerate cases fix the scale. If $C = \{\mathbf{c}\}$ is a single point, then $P_C(\mathbf{x}) = \mathbf{c}$ for every $\mathbf{x}$, and Eq. 17.3.1 holds because $\mathbf{y} - \mathbf{p} = \mathbf{0}$. If $C = V$, then P_C is the identity. In every case $P_C(\mathbf{x}) = \mathbf{x}$ exactly when $\mathbf{x} \in C$.

Quick check. Where did the proof of Theorem 17.3.1 use that C is closed, and where that it is convex? Match each use with one of the two failing experiments before the theorem.

Solution. Closedness was used in the existence step, to make K closed and hence compact; the open disc fails there, since the minimizing candidates converge to a point outside it. Convexity was used in $(\Rightarrow)$, to know that $(1 - t)\mathbf{p} + t\mathbf{y} \in C$, and uniqueness rests on that step. The circle fails there: with $\mathbf{x} = \mathbf{0}$, $\mathbf{p} = \mathbf{e}_1$ and $\mathbf{y} = -\mathbf{e}_1$, $\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle = 2 > 0$.

Warning. The proof needs an inner product, and uniqueness can fail without one. In $\mathbb{R}^2$ with $\|\cdot\|_\infty$, let C be the x_1 -axis, which is closed and convex, and $\mathbf{x} = (0, 1)$. Then $\|\mathbf{x} - (t, 0)\|_\infty = \max(|t|, 1) = 1$ for every $t \in [-1, 1]$, so there is a whole segment of nearest points. Existence survives, since its proof used only (N3) and compactness. Uniqueness does not, because it came from the expansion of $\|\mathbf{x} - \mathbf{y}\|^2$, which is available only for induced norms.

17.3.2 Hyperplanes and half-spaces

A hyperplane in the sense of Chapter 4 passes through $\mathbf{0}$: it is a subspace of codimension 1 (Definition 4.5.2). To separate sets we need translates of hyperplanes, and Chapter 4 §06 called a coset $\mathbf{p} + H$ of a hyperplane H (Definition 3.9.2) an **affine hyperplane**. In an inner product space these have a concrete form. By Theorem 4.5.3, $H = \ker \varphi$ for a non-zero $\varphi \in V^*$; by Theorem 10.5.1, $\varphi = \langle \cdot, \mathbf{a} \rangle$ for a non-zero $\mathbf{a} \in V$. Hence

$$\mathbf{p} + H = \{\mathbf{y} \in V : \langle \mathbf{a}, \mathbf{y} \rangle = c\}, \quad c = \langle \mathbf{a}, \mathbf{p} \rangle,$$

and conversely every such set with $\mathbf{a} \neq \mathbf{0}$ is an affine hyperplane, with $\mathbf{a}$ as a **normal vector**. It cuts V into the two **closed half-spaces** $\{\langle \mathbf{a}, \cdot \rangle \leq c\}$ and $\{\langle \mathbf{a}, \cdot \rangle \geq c\}$. Both are convex, because $\langle \mathbf{a}, \cdot \rangle$ is linear, and closed by Proposition 17.1.6 (c).

Definition 17.3.1 — Separation by a Hyperplane

Let $A, B \subseteq V$ be non-empty. A **non-zero** vector $\mathbf{a} \in V$ and a number $c \in \mathbb{R}$ **separate** A and B if

$$\langle \mathbf{a}, \mathbf{x} \rangle \leq c \leq \langle \mathbf{a}, \mathbf{y} \rangle \quad \text{for all } \mathbf{x} \in A \text{ and all } \mathbf{y} \in B.$$

They **strictly separate** A and B if in addition there is a real $\varepsilon > 0$ with $\langle \mathbf{a}, \mathbf{x} \rangle \leq c - \varepsilon$ and $\langle \mathbf{a}, \mathbf{y} \rangle \geq c + \varepsilon$ for all $\mathbf{x} \in A$, $\mathbf{y} \in B$.

In words: A lies in one closed half-space of $\{\langle \mathbf{a}, \cdot \rangle = c\}$ and B in the other, and strict separation puts a slab of width $2\varepsilon/\|\mathbf{a}\|$ between them. Replacing $(\mathbf{a}, c)$ by $(-\mathbf{a}, -c)$ swaps the roles of A and B . With $\mathbf{a} = \mathbf{0}$ and $c = 0$ any two sets would be “separated”.

17.3.3 Cutting off a point

The first consequence of [Theorem 17.3.1](#) is the one the chapter uses most. If $\mathbf{x}$ is not in C , the vector from $P_C(\mathbf{x})$ to $\mathbf{x}$ is a normal vector for a hyperplane that cuts $\mathbf{x}$ off.

Theorem 17.3.2 — Separating a Point from a Closed Convex Set

Let $C \subseteq V$ be non-empty, closed and convex, and let $\mathbf{x} \notin C$. Put $\mathbf{a} = \mathbf{x} - P_C(\mathbf{x})$. Then $\mathbf{a} \neq \mathbf{0}$ and

$$\langle \mathbf{a}, \mathbf{y} \rangle \leq \langle \mathbf{a}, \mathbf{x} \rangle - \|\mathbf{a}\|^2 \quad \text{for every } \mathbf{y} \in C.$$

In particular $\mathbf{a}$ and $c = \langle \mathbf{a}, \mathbf{x} \rangle - \frac{1}{2}\|\mathbf{a}\|^2$ strictly separate C and $\{\mathbf{x}\}$.

Proof. Write $\mathbf{p} = P_C(\mathbf{x})$. Since $\mathbf{p} \in C$ and $\mathbf{x} \notin C$, $\mathbf{a} = \mathbf{x} - \mathbf{p} \neq \mathbf{0}$. For $\mathbf{y} \in C$, [Eq. 17.3.1](#) gives $\langle \mathbf{a}, \mathbf{y} - \mathbf{p} \rangle \leq 0$, hence

$$\langle \mathbf{a}, \mathbf{y} \rangle \leq \langle \mathbf{a}, \mathbf{p} \rangle = \langle \mathbf{a}, \mathbf{x} \rangle - \langle \mathbf{a}, \mathbf{x} - \mathbf{p} \rangle = \langle \mathbf{a}, \mathbf{x} \rangle - \|\mathbf{a}\|^2.$$

With $\varepsilon = \frac{1}{2}\|\mathbf{a}\|^2 > 0$ this reads $\langle \mathbf{a}, \mathbf{y} \rangle \leq c - \varepsilon$, while $\langle \mathbf{a}, \mathbf{x} \rangle = c + \varepsilon$. This proves the theorem. ■

In the picture after [Theorem 17.3.1](#), the hyperplane of the theorem is the line parallel to the dashed one, halfway between $P_C(\mathbf{x})$ and $\mathbf{x}$.

The theorem turns into a description of a closed convex set from the outside. Sections 01 and 02 built convex sets up from points; this builds them down from half-spaces.

Corollary 17.3.3 — A Closed Convex Set Is an Intersection of Half-Spaces

Every non-empty closed convex subset C of V is the intersection of all the closed half-spaces $\{\mathbf{y} : \langle \mathbf{a}, \mathbf{y} \rangle \leq c\}$, with $\mathbf{a} \neq \mathbf{0}$, that contain C .

Proof. Each of these half-spaces contains C , so C lies in their intersection. Conversely, let $\mathbf{x} \notin C$. By [Theorem 17.3.2](#), with $\mathbf{a} = \mathbf{x} - P_C(\mathbf{x}) \neq \mathbf{0}$, the half-space $\{\mathbf{y} : \langle \mathbf{a}, \mathbf{y} \rangle \leq \langle \mathbf{a}, \mathbf{x} \rangle - \|\mathbf{a}\|^2\}$ contains C but not $\mathbf{x}$. So $\mathbf{x}$ is not in the intersection. (If $C = V$, the family of half-spaces is empty and its intersection is V by convention.) ■

The corollary has a converse that is immediate: an intersection of closed half-spaces is closed and convex, because each half-space is and both properties pass to intersections. So the closed convex sets are **exactly** the intersections of closed half-spaces. §04 turns this into a formula, the support function, and §09 studies the case of finitely many half-spaces.

17.3.4 Supporting hyperplanes

A point on the edge of C cannot be cut off, because it belongs to C . The best one can hope for is a hyperplane through the point with all of C on one side, touching C without cutting it. To say “on the edge” we need two more words.

Definition 17.3.2 — Interior Point, Closure, Boundary Point

Let $S \subseteq V$. A point $\mathbf{x} \in V$ is an **interior point** of S if there is a real $r > 0$ such that **every** $\mathbf{y}$ with $\|\mathbf{y} - \mathbf{x}\| < r$ lies in S ; the set of interior points is written $\text{int}(S)$. The **closure** $\bar{S}$ of S is the set of limits of all convergent sequences of points of S . A **boundary point** of S is a point of $\bar{S}$ that is not an interior point of S .

Constant sequences show $S \subseteq \bar{S}$, and by Definition 17.1.6, S is closed exactly when $\bar{S} = S$. By Theorem 15.2.2 neither notion depends on the norm used: a ball of radius r for one norm contains a ball of positive radius for any other. For the closed unit disc of $\mathbb{R}^2$, the interior points are those with $\|\mathbf{x}\| < 1$, and the boundary points form the unit circle. A segment in $\mathbb{R}^2$ has no interior points at all, so every one of its points is a boundary point; that is the case the next lemma isolates.

Lemma 17.3.4 — A Convex Set Is Solid or Flat

Let $V \neq \{\mathbf{0}\}$ and let $C \subseteq V$ be non-empty and convex. Then **either** C has an interior point, **or** C lies in an affine hyperplane: there are $\mathbf{a} \neq \mathbf{0}$ and $c \in \mathbb{R}$ with $\langle \mathbf{a}, \mathbf{y} \rangle = c$ for every $\mathbf{y} \in C$.

Idea. Fix $\mathbf{c}_0 \in C$ and look at the directions $\mathbf{y} - \mathbf{c}_0$ available inside C . If they span a proper subspace, a normal vector to that subspace is the $\mathbf{a}$ we want. If they span V , then C contains $n + 1$ points in general position, hence the solid simplex they span, and the center of a solid simplex is an interior point because its barycentric coordinates change only a little when the point moves a little.

Proof. Fix $\mathbf{c}_0 \in C$ and let $W = \text{span}\{\mathbf{y} - \mathbf{c}_0 : \mathbf{y} \in C\}$.

Case 1: $W \neq V$. Then $\dim W < n$, so $\dim W^\perp = n - \dim W \geq 1$ by Theorem 10.3.2 (c), and there is $\mathbf{a} \neq \mathbf{0}$ in $W^\perp$. For $\mathbf{y} \in C$, $\langle \mathbf{a}, \mathbf{y} - \mathbf{c}_0 \rangle = 0$, that is, $\langle \mathbf{a}, \mathbf{y} \rangle = \langle \mathbf{a}, \mathbf{c}_0 \rangle$. Take $c = \langle \mathbf{a}, \mathbf{c}_0 \rangle$.

Case 2: $W = V$. Choose a linearly independent list $(\mathbf{y}_1 - \mathbf{c}_0, \dots, \mathbf{y}_m - \mathbf{c}_0)$ with every $\mathbf{y}_i \in C$ and m as large as possible; $m \leq n$ by Theorem 1.6.3. Every vector $\mathbf{y} - \mathbf{c}_0$ with $\mathbf{y} \in C$ lies in the span of this list, since otherwise Lemma 1.5.3 would lengthen it. So the span contains every generator of W , hence equals V , and the list is a basis; thus $m = n$. Let $\varphi_i(\mathbf{v})$ be the i -th coordinate of $\mathbf{v}$ in this basis, a linear functional by Theorem 1.5.2, and write $\varphi_i = \langle \cdot, \mathbf{w}_i \rangle$ by Theorem 10.5.1. Put $\mu_i(\mathbf{v}) = \varphi_i(\mathbf{v} - \mathbf{c}_0)$. Then for every $\mathbf{v} \in V$,

$$\mathbf{v} = \left(1 - \sum_{i=1}^n \mu_i(\mathbf{v})\right) \mathbf{c}_0 + \sum_{i=1}^n \mu_i(\mathbf{v}) \mathbf{y}_i,$$

and when $\mu_i(\mathbf{v}) \geq 0$ for every i and $\sum_i \mu_i(\mathbf{v}) \leq 1$, this is a convex combination of points of C , hence a point of C : by Theorem 17.1.3 it lies in $\text{conv } C$, and $\text{conv } C = C$ by

Definition 17.1.4, since C is itself a convex set containing C .

Let $\mathbf{z} = \mathbf{c}_0 + \frac{1}{n+1} \sum_i (\mathbf{y}_i - \mathbf{c}_0)$, so that $\mu_i(\mathbf{z}) = \frac{1}{n+1}$ for every i . Each $\mathbf{w}_i \neq \mathbf{0}$, because $\varphi_i(\mathbf{y}_i - \mathbf{c}_0) = 1$; put $L = \max_i \|\mathbf{w}_i\| > 0$ and $r = 1/(n(n+1)L)$. If $\|\mathbf{v} - \mathbf{z}\| < r$, then by Theorem 10.1.5

$$|\mu_i(\mathbf{v}) - \mu_i(\mathbf{z})| = |\langle \mathbf{v} - \mathbf{z}, \mathbf{w}_i \rangle| \leq L \|\mathbf{v} - \mathbf{z}\| < \frac{1}{n(n+1)},$$

so $\mu_i(\mathbf{v}) > \frac{1}{n+1} - \frac{1}{n(n+1)} \geq 0$ and $\sum_i \mu_i(\mathbf{v}) < \frac{n}{n+1} + \frac{1}{n+1} = 1$. Hence $\mathbf{v} \in C$, and $\mathbf{z}$ is an interior point of C . ■

The second ingredient says that from an interior point one can see the whole closure.

Lemma 17.3.5 — Closure and Interior of a Convex Set

Let $C \subseteq V$ be convex.

- (a) $\overline{C}$ is closed and convex.
- (b) If $\mathbf{z} \in \text{int}(C)$, $\mathbf{w} \in \overline{C}$ and $0 < t \leq 1$, then $(1-t)\mathbf{w} + t\mathbf{z} \in \text{int}(C)$.

Idea. Part (a) pushes limits through: a limit of limits is a limit, and a convex combination of limits is the limit of convex combinations. For (b), picture a small ball around $\mathbf{z}$ inside C , and the cone from $\mathbf{w}$ over it. The cone contains a smaller ball around $(1-t)\mathbf{w} + t\mathbf{z}$; since $\mathbf{w}$ is only a limit, replace it by a nearby point $\mathbf{w}'$ of C , which shrinks the ball by half.

Proof.

- (a) Let $\mathbf{x}_k \in \overline{C}$ with $\mathbf{x}_k \rightarrow \mathbf{x}$. Each $\mathbf{x}_k$ is a limit of points of C , so there is $\mathbf{c}_k \in C$ with $\|\mathbf{c}_k - \mathbf{x}_k\| < 1/k$. Then $\|\mathbf{c}_k - \mathbf{x}\| \leq 1/k + \|\mathbf{x}_k - \mathbf{x}\| \rightarrow 0$, so $\mathbf{x} \in \overline{C}$. For convexity, let $\mathbf{x} = \lim \mathbf{c}_k$ and $\mathbf{y} = \lim \mathbf{d}_k$ with $\mathbf{c}_k, \mathbf{d}_k \in C$, and $0 \leq s \leq 1$. The points $s\mathbf{c}_k + (1-s)\mathbf{d}_k$ lie in C and converge to $s\mathbf{x} + (1-s)\mathbf{y}$, by Corollary 10.1.6, so that point lies in $\overline{C}$.
- (b) Take $\rho > 0$ with every point within distance ρ of $\mathbf{z}$ in C , and put $\mathbf{p} = (1-t)\mathbf{w} + t\mathbf{z}$. If $t = 1$ then $\mathbf{p} = \mathbf{z}$ and there is nothing to prove, so let $t < 1$. Choose $\mathbf{w}' \in C$ with $(1-t)\|\mathbf{w} - \mathbf{w}'\| < t\rho/2$, possible because $\mathbf{w}$ is a limit of points of C . Let $\|\mathbf{q} - \mathbf{p}\| < t\rho/2$, and put $\mathbf{z}' = (\mathbf{q} - (1-t)\mathbf{w}')/t$, so that $\mathbf{q} = (1-t)\mathbf{w}' + t\mathbf{z}'$. Since $\mathbf{p} = (1-t)\mathbf{w} + t\mathbf{z}$,

$$\mathbf{z}' - \mathbf{z} = \frac{(\mathbf{q} - \mathbf{p}) + (1-t)(\mathbf{w} - \mathbf{w}')}{t}, \quad \|\mathbf{z}' - \mathbf{z}\| < \frac{t\rho/2 + t\rho/2}{t} = \rho.$$

Hence $\mathbf{z}' \in C$, and $\mathbf{q} \in C$ by Definition 17.1.1. So every point within $t\rho/2$ of $\mathbf{p}$ lies in C , and $\mathbf{p} \in \text{int}(C)$. ■

Now the theorem. It asks only that $\mathbf{x}_0$ not be an interior point, which covers the boundary points of C and also the points outside $\overline{C}$.

Theorem 17.3.6 — Supporting Hyperplane Theorem

Let $V \neq \{0\}$, let $C \subseteq V$ be convex, and let $\mathbf{x}_0 \in V$ be a point that is **not** an interior point of C . Then there is a **non-zero** $\mathbf{a} \in V$ with

$$\langle \mathbf{a}, \mathbf{y} \rangle \leq \langle \mathbf{a}, \mathbf{x}_0 \rangle \quad \text{for every } \mathbf{y} \in C.$$

If moreover $\mathbf{x}_0 \in C$, the affine hyperplane $\{\mathbf{y} : \langle \mathbf{a}, \mathbf{y} \rangle = \langle \mathbf{a}, \mathbf{x}_0 \rangle\}$ passes through $\mathbf{x}_0$ and has C in one of its closed half-spaces; it is called a **supporting hyperplane** of C at $\mathbf{x}_0$.

Idea. Approach $\mathbf{x}_0$ from outside $\overline{C}$ and cut off each approximating point with [Theorem 17.3.2](#). The normal vectors, scaled to length 1, live on the compact unit sphere, so a subsequence converges, and in the limit the strict inequalities become the non-strict one we want. The only thing that could go wrong is that there are no points outside $\overline{C}$ near $\mathbf{x}_0$. [Lemma 17.3.4](#) and [Lemma 17.3.5](#) rule that out: if C is flat, its own hyperplane serves; if C is solid, pushing $\mathbf{x}_0$ directly away from an interior point leaves $\overline{C}$ at once.

Proof. If $C = \emptyset$, any non-zero $\mathbf{a}$ works, and one exists because $V \neq \{0\}$. So let $C \neq \emptyset$.

Case 1: C has no interior point. By [Lemma 17.3.4](#) there are $\mathbf{a} \neq \mathbf{0}$ and c with $\langle \mathbf{a}, \mathbf{y} \rangle = c$ for all $\mathbf{y} \in C$. If $\langle \mathbf{a}, \mathbf{x}_0 \rangle \geq c$, then $\mathbf{a}$ works. Otherwise $-\mathbf{a}$ works, since $\langle -\mathbf{a}, \mathbf{y} \rangle = -c < -\langle \mathbf{a}, \mathbf{x}_0 \rangle$.

Case 2: C has an interior point $\mathbf{z}$. For $k \geq 1$ let $\mathbf{x}_k = \mathbf{x}_0 + \frac{1}{k}(\mathbf{x}_0 - \mathbf{z})$. We claim $\mathbf{x}_k \notin \overline{C}$. Indeed, with $t = \frac{1}{k+1}$,

$$(1-t)\mathbf{x}_k + t\mathbf{z} = \frac{k}{k+1}\mathbf{x}_0 + \frac{1}{k+1}(\mathbf{x}_0 - \mathbf{z}) + \frac{1}{k+1}\mathbf{z} = \mathbf{x}_0,$$

so if $\mathbf{x}_k$ were in $\overline{C}$, then $\mathbf{x}_0$ would be an interior point of C by [Lemma 17.3.5](#) (b), contrary to the hypothesis. By [Lemma 17.3.5](#) (a), $\overline{C}$ is non-empty, closed and convex, so [Theorem 17.3.2](#), applied to $\overline{C}$ and $\mathbf{x}_k$, gives $\mathbf{b}_k \neq \mathbf{0}$ with $\langle \mathbf{b}_k, \mathbf{y} \rangle \leq \langle \mathbf{b}_k, \mathbf{x}_k \rangle$ for all $\mathbf{y} \in \overline{C}$, in particular for all $\mathbf{y} \in C$. Put $\mathbf{a}_k = \mathbf{b}_k / \|\mathbf{b}_k\|$, a unit vector satisfying the same inequality.

The unit sphere of V is compact by Corollary 15.2.5, which rests on **fact (A3) of Chapter 15's introduction**. So a subsequence $(\mathbf{a}_{k_j})$ converges to some $\mathbf{a}$ with $\|\mathbf{a}\| = 1$; in particular $\mathbf{a} \neq \mathbf{0}$. Fix $\mathbf{y} \in C$. Then $\langle \mathbf{a}_{k_j}, \mathbf{y} - \mathbf{x}_{k_j} \rangle \leq 0$ for every j , and $\mathbf{x}_k \rightarrow \mathbf{x}_0$ because $\|\mathbf{x}_k - \mathbf{x}_0\| = \|\mathbf{x}_0 - \mathbf{z}\|/k$. By Theorem 10.1.5,

$$\begin{aligned} |\langle \mathbf{a}_{k_j}, \mathbf{y} - \mathbf{x}_{k_j} \rangle - \langle \mathbf{a}, \mathbf{y} - \mathbf{x}_0 \rangle| &\leq |\langle \mathbf{a}_{k_j} - \mathbf{a}, \mathbf{y} - \mathbf{x}_{k_j} \rangle| + |\langle \mathbf{a}, \mathbf{x}_0 - \mathbf{x}_{k_j} \rangle| \\ &\leq \|\mathbf{a}_{k_j} - \mathbf{a}\|(\|\mathbf{y} - \mathbf{x}_0\| + \|\mathbf{x}_0 - \mathbf{z}\|) + \|\mathbf{x}_0 - \mathbf{x}_{k_j}\|, \end{aligned}$$

which tends to 0. A non-strict inequality survives the limit, so $\langle \mathbf{a}, \mathbf{y} - \mathbf{x}_0 \rangle \leq 0$. This proves the theorem. ■

A supporting hyperplane need not be unique. At the corner $(1, 1)$ of the square $[-1, 1]^2$, every $\mathbf{a} = (a_1, a_2)$ with $a_1, a_2 \geq 0$, not both zero, works, since $a_1 y_1 + a_2 y_2 \leq a_1 + a_2$ on the square. At a point of the unit circle, the boundary of the disc, only the positive multiples of the point itself work. Corners admit many supporting hyperplanes and smooth points one; §08 builds

its induction on dimension on these hyperplanes, and §04 uses them to prove that a norm is recovered from its dual.

17.3.5 Separating two convex sets

To separate two sets, separate one point from one set: the point $\mathbf{0}$ from the set of differences.

Theorem 17.3.7 — Separating Hyperplane Theorem

Let $C_1, C_2 \subseteq V$ be **non-empty, disjoint** and convex.

- (a) If C_1 is **compact** and C_2 is **closed**, then C_1 and C_2 are strictly separated: there are $\mathbf{a} \neq \mathbf{0}$ and c with $\langle \mathbf{a}, \mathbf{x} \rangle < c < \langle \mathbf{a}, \mathbf{y} \rangle$ for all $\mathbf{x} \in C_1, \mathbf{y} \in C_2$, and indeed with a margin $\varepsilon > 0$ as in Definition 17.3.1.
- (b) In general, C_1 and C_2 are separated: there are $\mathbf{a} \neq \mathbf{0}$ and c with $\langle \mathbf{a}, \mathbf{x} \rangle \leq c \leq \langle \mathbf{a}, \mathbf{y} \rangle$ for all $\mathbf{x} \in C_1, \mathbf{y} \in C_2$.

Idea. The difference set $D = \{\mathbf{y} - \mathbf{x} : \mathbf{x} \in C_1, \mathbf{y} \in C_2\}$ is convex and misses $\mathbf{0}$, and a normal $\mathbf{a}$ with D on its positive side separates C_1 from C_2 . For (a), compactness of C_1 makes D closed, and Theorem 17.3.2 cuts $\mathbf{0}$ off; for (b), $\mathbf{0}$ is not an interior point of D , and Theorem 17.3.6 applies.

Proof. Let $D = \{\mathbf{y} - \mathbf{x} : \mathbf{x} \in C_1, \mathbf{y} \in C_2\}$. It is convex: for $0 \leq t \leq 1$,

$$t(\mathbf{y} - \mathbf{x}) + (1 - t)(\mathbf{y}' - \mathbf{x}') = (t\mathbf{y} + (1 - t)\mathbf{y}') - (t\mathbf{x} + (1 - t)\mathbf{x}'),$$

a difference of a point of C_2 and a point of C_1 . And $\mathbf{0} \notin D$, since $C_1 \cap C_2 = \emptyset$.

- (a) We show D is closed. Let $\mathbf{y}_k - \mathbf{x}_k \rightarrow \mathbf{d}$ with $\mathbf{x}_k \in C_1, \mathbf{y}_k \in C_2$. Since C_1 is compact, a subsequence $\mathbf{x}_{k_j}$ converges to some $\mathbf{x} \in C_1$. Then $\mathbf{y}_{k_j} = (\mathbf{y}_{k_j} - \mathbf{x}_{k_j}) + \mathbf{x}_{k_j} \rightarrow \mathbf{d} + \mathbf{x}$, since a subsequence of a convergent sequence has the same limit, part of the elementary algebra of limits that Chapter 15's introduction imports; so $\mathbf{d} + \mathbf{x} \in C_2$ because C_2 is closed, and $\mathbf{d} = (\mathbf{d} + \mathbf{x}) - \mathbf{x} \in D$. Now Theorem 17.3.2, applied to D and the point $\mathbf{0} \notin D$, gives $\mathbf{b} \neq \mathbf{0}$ with $\langle \mathbf{b}, \mathbf{d} \rangle \leq -\|\mathbf{b}\|^2$ for all $\mathbf{d} \in D$. With $\mathbf{a} = -\mathbf{b}$ this says

$$\langle \mathbf{a}, \mathbf{x} \rangle + \|\mathbf{a}\|^2 \leq \langle \mathbf{a}, \mathbf{y} \rangle \quad \text{for all } \mathbf{x} \in C_1, \mathbf{y} \in C_2.$$

Fix $\mathbf{y} \in C_2$. The numbers $\langle \mathbf{a}, \mathbf{x} \rangle, \mathbf{x} \in C_1$, are bounded above by $\langle \mathbf{a}, \mathbf{y} \rangle - \|\mathbf{a}\|^2$, so by the completeness of $\mathbb{R}$, fact (A1) of Chapter 15's introduction, they have a least upper bound s , and $s + \|\mathbf{a}\|^2 \leq \langle \mathbf{a}, \mathbf{y} \rangle$ for every $\mathbf{y} \in C_2$. Take $\varepsilon = \frac{1}{2}\|\mathbf{a}\|^2$ and $c = s + \varepsilon$. Then $\langle \mathbf{a}, \mathbf{x} \rangle \leq s = c - \varepsilon$ and $\langle \mathbf{a}, \mathbf{y} \rangle \geq c + \varepsilon$.

- (b) If $V = \{\mathbf{0}\}$, two non-empty subsets both contain $\mathbf{0}$ and cannot be disjoint, so $V \neq \{\mathbf{0}\}$. The point $\mathbf{0}$ is not in D , so it is not an interior point of D . By Theorem 17.3.6 there is $\mathbf{b} \neq \mathbf{0}$ with $\langle \mathbf{b}, \mathbf{d} \rangle \leq \langle \mathbf{b}, \mathbf{0} \rangle = 0$ for all $\mathbf{d} \in D$. With $\mathbf{a} = -\mathbf{b}$, this says $\langle \mathbf{a}, \mathbf{x} \rangle \leq \langle \mathbf{a}, \mathbf{y} \rangle$ for all $\mathbf{x} \in C_1, \mathbf{y} \in C_2$. As in (a), the least upper bound c of the numbers $\langle \mathbf{a}, \mathbf{x} \rangle, \mathbf{x} \in C_1$, exists by (A1) and satisfies $\langle \mathbf{a}, \mathbf{x} \rangle \leq c \leq \langle \mathbf{a}, \mathbf{y} \rangle$. This proves the theorem.

17.3.6 Where strict separation fails

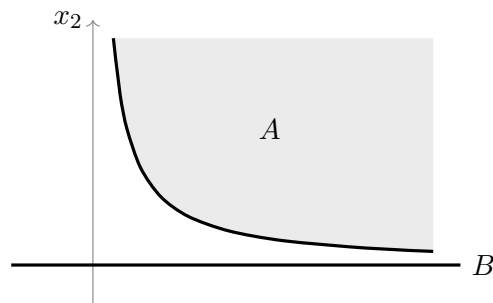
Part (a) asked one of the two sets to be compact. That hypothesis cannot be weakened to “closed”, and the standard witness is a hyperbola and its asymptote.

Example. (A hyperbola and its asymptote)

In $\mathbb{R}^2$ with the dot product, let

$$A = \{\mathbf{x} : x_1 > 0, x_2 > 0, x_1x_2 \geq 1\}, \quad B = \{\mathbf{x} : x_2 = 0\}.$$

Show that A and B are non-empty, disjoint, closed and convex, that the x_1 -axis is the **only** hyperplane separating them, and that they are **not** strictly separated.



A and B are closed, convex and disjoint, but the gap between them shrinks to zero

Solution. Non-empty and disjoint. $(1, 1) \in A$ and $\mathbf{0} \in B$; every point of A has $x_2 > 0$, and every point of B has $x_2 = 0$.

B is closed and convex. It is a subspace, so it is convex, and closed by Proposition 17.1.6 (c).

A is convex. Let $\mathbf{x}, \mathbf{z} \in A$ and $0 \leq t \leq 1$, and let $\mathbf{w} = t\mathbf{x} + (1-t)\mathbf{z}$. Its coordinates are positive. Expanding,

$$w_1w_2 = t^2x_1x_2 + (1-t)^2z_1z_2 + t(1-t)(x_1z_2 + x_2z_1).$$

By $(\sqrt{x_1z_2} - \sqrt{x_2z_1})^2 \geq 0$, we get $x_1z_2 + x_2z_1 \geq 2\sqrt{x_1x_2z_1z_2} \geq 2$, using $x_1x_2 \geq 1$ and $z_1z_2 \geq 1$. Hence $w_1w_2 \geq t^2 + (1-t)^2 + 2t(1-t) = 1$, and $\mathbf{w} \in A$.

A is closed. If $\mathbf{x}_k \rightarrow \mathbf{x}$ with $\mathbf{x}_k \in A$, then the coordinates converge, so $x_1, x_2 \geq 0$, and $x_{k,1}x_{k,2} \rightarrow x_1x_2$ by the algebra of limits, so $x_1x_2 \geq 1$. A product that is at least 1 has no zero factor, so $x_1, x_2 > 0$, and $\mathbf{x} \in A$.

The only separating hyperplane. Let $\mathbf{a} \neq \mathbf{0}$ and c separate A and B , in either order. On B , $\langle \mathbf{a}, (t, 0) \rangle = a_1t$ for all real t , and these values lie on one side of c ; that forces $a_1 = 0$. So $a_2 \neq 0$, and after replacing $(\mathbf{a}, c)$ by $(-\mathbf{a}, -c)$ we may take $a_2 > 0$. On B the value is 0. On A the values are a_2x_2 , and since $(1/s, s) \in A$ for every $s > 0$, they fill the interval $(0, \infty)$. These are unbounded above, so A is on the side $\geq c$ and B on the side

$\leq c$: $0 \leq c \leq a_2 s$ for every $s > 0$, which forces $c = 0$. The hyperplane $\{a_2 x_2 = 0\}$ is the x_1 -axis, and it separates, since $a_2 x_2 > 0$ on A and $= 0$ on B .

No strict separation. Strict separation would need the only separating hyperplane to miss both sets, and B is that hyperplane.

The points $(t, 1/t) \in A$ and $(t, 0) \in B$ are $1/t$ apart, and [Exercise \(C2: Strict separation and distance\)](#) shows that this vanishing distance is exactly what rules out strict separation.

Warning. **Disjoint closed convex sets need not be strictly separated.** The hyperbola region and its asymptote above are the example to remember: both closed, both convex, disjoint, and every hyperplane that separates them contains one of them. “Disjoint” gives only the weak separation of [Theorem 17.3.7](#) (b). Strict separation needs more, such as compactness of one set in (a).

17.3.7 Extending linear functionals

Separation has an analytic twin. Instead of a set and a point, take a linear functional defined on a subspace and bounded there by a convex function; can it be extended to the whole space without breaking the bound? The bounds allowed are the following.

Definition 17.3.3 — Sublinear Functional

Let V be a real vector space. A function $p: V \rightarrow \mathbb{R}$ is **sublinear** if, for **all** $\mathbf{x}, \mathbf{y} \in V$ and **all** real $t \geq 0$,

$$p(\mathbf{x} + \mathbf{y}) \leq p(\mathbf{x}) + p(\mathbf{y}) \quad \text{and} \quad p(t\mathbf{x}) = t p(\mathbf{x}).$$

Every norm is sublinear: (N3) is the first condition, and (N2) with $t \geq 0$ the second. Every linear functional is sublinear, with equality in the first condition. So is $p(\mathbf{x}) = \max(x_1, x_2)$ on $\mathbb{R}^2$, which is neither a norm nor linear: $p(-\mathbf{e}_1) = 0$. Taking $t = 0$ gives $p(\mathbf{0}) = 0$. A non-example by minimal change is $p(\mathbf{x}) = \|\mathbf{x}\|^2$: the first condition fails at $\mathbf{x} = \mathbf{y} \neq \mathbf{0}$ (it would give $4\|\mathbf{x}\|^2 \leq 2\|\mathbf{x}\|^2$), and so does the second, since $p(t\mathbf{x}) = t^2 p(\mathbf{x})$.

It turns out that sublinearity is exactly what makes one-step extensions possible.

Theorem 17.3.8 — Hahn–Banach Theorem, Finite-Dimensional

Let V be a **finite-dimensional** real vector space, let p be a sublinear functional on V , let U be a subspace of V , and let $f \in U^*$ satisfy $f(\mathbf{u}) \leq p(\mathbf{u})$ for every $\mathbf{u} \in U$. Then there is $F \in V^*$ with

$$F(\mathbf{u}) = f(\mathbf{u}) \text{ for every } \mathbf{u} \in U \quad \text{and} \quad F(\mathbf{v}) \leq p(\mathbf{v}) \text{ for every } \mathbf{v} \in V.$$

Idea. Add one dimension at a time. On $U + \text{span}(\mathbf{v})$, a linear extension is fixed by one number $c = F(\mathbf{v})$. The requirement $F \leq p$ at the vectors $\mathbf{u} + t\mathbf{v}$ with $t > 0$ gives upper bounds on c , and at those with $t < 0$ it gives lower bounds. Sublinearity of p , together with linearity of f , is exactly what makes every lower bound at most every upper bound, so some c fits. Then induct on $\dim V - \dim U$. No inner product is needed.

Proof. We induct on $k = \dim V - \dim U$. If $k = 0$, then $U = V$ and $F = f$ works.

Let $k \geq 1$ and assume the theorem whenever the difference of dimensions is $k - 1$. Pick $\mathbf{v} \in V \setminus U$ and let $W = U + \text{span}(\mathbf{v})$. Each $\mathbf{w} \in W$ can be written as $\mathbf{u} + t\mathbf{v}$ with $\mathbf{u} \in U$ and $t \in \mathbb{R}$, and only in one way: if $\mathbf{u} + t\mathbf{v} = \mathbf{u}' + t'\mathbf{v}$ with $t \neq t'$, then $\mathbf{v} = (\mathbf{u}' - \mathbf{u})/(t - t') \in U$, which is false; so $t = t'$ and $\mathbf{u} = \mathbf{u}'$. Appending $\mathbf{v}$ to a basis of U gives a basis of W by Lemma 1.5.3, so $\dim V - \dim W = k - 1$. For $c \in \mathbb{R}$, the formula $g(\mathbf{u} + t\mathbf{v}) = f(\mathbf{u}) + tc$ therefore defines a linear functional on W that extends f .

For $\mathbf{u}, \mathbf{u}' \in U$, linearity of f , the hypothesis, and sublinearity of p give

$$f(\mathbf{u}) + f(\mathbf{u}') = f(\mathbf{u} + \mathbf{u}') \leq p(\mathbf{u} + \mathbf{u}') \leq p(\mathbf{u} - \mathbf{v}) + p(\mathbf{u}' + \mathbf{v}),$$

that is, $f(\mathbf{u}) - p(\mathbf{u} - \mathbf{v}) \leq p(\mathbf{u}' + \mathbf{v}) - f(\mathbf{u}')$. So every number in $L = \{f(\mathbf{u}) - p(\mathbf{u} - \mathbf{v}) : \mathbf{u} \in U\}$ is at most every number in $R = \{p(\mathbf{u}' + \mathbf{v}) - f(\mathbf{u}') : \mathbf{u}' \in U\}$. By **fact (A1) of Chapter 15's introduction**, the non-empty set L , bounded above by any element of R , has a least upper bound c , and $c \leq r$ for every $r \in R$. With this c , we check $g(\mathbf{u} + t\mathbf{v}) \leq p(\mathbf{u} + t\mathbf{v})$ in three cases.

- $t = 0$: this is the hypothesis $f(\mathbf{u}) \leq p(\mathbf{u})$.
- $t > 0$: $c \leq p(\mathbf{u}/t + \mathbf{v}) - f(\mathbf{u}/t)$, the element of R at $\mathbf{u}' = \mathbf{u}/t$. Multiply by t and use $p(t\cdot) = tp(\cdot)$: $tc \leq p(\mathbf{u} + t\mathbf{v}) - f(\mathbf{u})$.
- $t = -s < 0$: $c \geq f(\mathbf{u}/s) - p(\mathbf{u}/s - \mathbf{v})$, since this is an element of L . Multiply by s : $sc \geq f(\mathbf{u}) - p(\mathbf{u} - s\mathbf{v})$, that is, $f(\mathbf{u}) - sc \leq p(\mathbf{u} + t\mathbf{v})$.

So $g \leq p$ on W . By the induction hypothesis applied to W and g , there is $F \in V^*$ extending g , hence f , with $F \leq p$ on V . This proves the theorem. ■

Beyond finite dimension. The induction above stops after $\dim V - \dim U$ steps. In an infinite-dimensional space there is no last step, and the one-step extension has to be combined with Zorn's lemma (Theorem 1.8.1), exactly as Chapter 1 §08 combined “append one vector” with Zorn's lemma to prove Theorem 1.8.3. The resulting theorem, the Hahn–Banach theorem of functional analysis, is outside the scope of this book, and nothing here depends on it.

The special case that matters most is a norm.

Corollary 17.3.9 — Norm-Preserving Extension

Let V be a finite-dimensional real vector space with a norm $\|\cdot\|$, let U be a subspace, and let $f \in U^*$ satisfy $|f(\mathbf{u})| \leq \|\mathbf{u}\|$ for every $\mathbf{u} \in U$. Then there is $F \in V^*$ extending f with $|F(\mathbf{v})| \leq \|\mathbf{v}\|$ for every $\mathbf{v} \in V$.

Proof. A norm is sublinear, so [Theorem 17.3.8](#) with $p = \|\cdot\|$ gives F extending f with $F(\mathbf{v}) \leq \|\mathbf{v}\|$ for all $\mathbf{v}$. Applying this to $-\mathbf{v}$ gives $-F(\mathbf{v}) = F(-\mathbf{v}) \leq \|-\mathbf{v}\| = \|\mathbf{v}\|$ by (N2). Hence $|F(\mathbf{v})| \leq \|\mathbf{v}\|$. ■

In particular, for any $\mathbf{x} \neq \mathbf{0}$, taking $U = \text{span}(\mathbf{x})$ and $f(t\mathbf{x}) = t\|\mathbf{x}\|$ produces a functional F with $F(\mathbf{x}) = \|\mathbf{x}\|$ and $|F| \leq \|\cdot\|$ everywhere: a linear functional that “sees” the full length of $\mathbf{x}$

without ever exceeding the norm. §04 meets this functional again, as the vector at which a dual norm is attained.

Example. (Many extensions)

On $\mathbb{R}^2$ with $\|\cdot\|_\infty$, let $U = \text{span}((1, 1))$ and $f(s(1, 1)) = s$. Find every F as in [Corollary 17.3.9](#), and compare with the interval produced by the proof of [Theorem 17.3.8](#) for $\mathbf{v} = \mathbf{e}_1$.

Solution. First, $|f(s(1, 1))| = |s| = \|s(1, 1)\|_\infty$, so the hypothesis holds. An extension has $F(\mathbf{e}_1) = c$ for some c , and then $F(\mathbf{e}_2) = F((1, 1)) - c = 1 - c$, so $F(\mathbf{x}) = cx_1 + (1 - c)x_2$.

If $0 \leq c \leq 1$, then $|F(\mathbf{x})| \leq c|x_1| + (1 - c)|x_2| \leq \|\mathbf{x}\|_\infty$. If $c > 1$, then $\mathbf{x} = (1, -1)$ gives $F(\mathbf{x}) = 2c - 1 > 1 = \|\mathbf{x}\|_\infty$. If $c < 0$, then $\mathbf{x} = (-1, 1)$ gives $F(\mathbf{x}) = 1 - 2c > 1$. So the extensions are exactly $F(\mathbf{x}) = cx_1 + (1 - c)x_2$ with $c \in [0, 1]$: infinitely many.

In the proof, with $p = \|\cdot\|_\infty$, $\mathbf{u} = s(1, 1)$ and $\mathbf{v} = \mathbf{e}_1$, the elements of L are $s - \max(|s - 1|, |s|)$, which is 0 for $s \geq \frac{1}{2}$ and $2s - 1 < 0$ for $s < \frac{1}{2}$; so $\sup L = 0$. The elements of R are $\max(|s + 1|, |s|) - s$, which is 1 for $s \geq -\frac{1}{2}$ and $-2s > 1$ for $s < -\frac{1}{2}$; so $\inf R = 1$. Every c between $\sup L$ and $\inf R$ works, and that is the interval $[0, 1]$ found directly. The proof picked its left end.

The corners of the square B_∞ are why the extension is not unique: U meets the unit sphere at the corner $(1, 1)$, and every supporting line of the square at that corner gives an extension.

17.3.8 Exercises

17.3.8.1 A. Check your understanding

Exercise A1.

- State the characterization [Eq. 17.3.1](#) of the nearest point, and say what it becomes when C is a subspace.
- Which two facts of Chapter 15's introduction does the existence of $P_C(\mathbf{x})$ use, and to which set are they applied?
- Determine whether the following statement is correct, and justify your answer: every non-empty closed subset of $\mathbb{R}^2$ has a unique point nearest to $\mathbf{0}$.
- Determine whether the following statement is correct, and justify your answer: two disjoint non-empty closed convex subsets of $\mathbb{R}^2$ can always be strictly separated.

Solution.

- A point $\mathbf{p} \in C$ is $P_C(\mathbf{x})$ exactly when $\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle \leq 0$ for every $\mathbf{y} \in C$ ([Theorem 17.3.1](#) (b)). For a subspace U it becomes the equality $\langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle = 0$ for every $\mathbf{u} \in U$, so $P_C(\mathbf{x}) = P_U \mathbf{x}$ and [Theorem 10.3.5](#) is recovered.
- The compactness of closed bounded sets (A3), through [Corollary 15.2.5](#), and the extreme value theorem (A4). They are applied to $K = \{\mathbf{y} \in C : \|\mathbf{x} - \mathbf{y}\| \leq R\}$, the part of C within the distance R of $\mathbf{x}$ attained at one chosen point of C , and to the function $\mathbf{y} \mapsto \|\mathbf{x} - \mathbf{y}\|$.

- (c) Incorrect. The unit circle is closed and non-empty, and every one of its points is at distance 1 from $\mathbf{0}$. Uniqueness needs convexity.
- (d) Incorrect. The region A above the hyperbola $x_1x_2 = 1$ in the positive quadrant and its asymptote B , the x_1 -axis, are disjoint, closed and convex, but the only hyperplane separating them is B itself (Example (A hyperbola and its asymptote)).

17.3.8.2 B. Practice

Exercise B1: Projection onto a half-plane.

In $\mathbb{R}^2$ with the dot product, let $C = \{\mathbf{y} : y_1 + y_2 \leq 1\}$ and $\mathbf{x} = (2, 3)$. Find $P_C(\mathbf{x})$ and $d(\mathbf{x}, C)$, and verify Eq. 17.3.1.

Solution. Here $\mathbf{a} = (1, 1)$, $c = 1$, and $\langle \mathbf{a}, \mathbf{x} \rangle = 5 > 1$. By Example (Three projections) (c), $\lambda = (5 - 1)/2 = 2$ and $P_C(\mathbf{x}) = (2, 3) - 2(1, 1) = (0, 1)$. Check: $0 + 1 = 1$, so $\mathbf{p} = (0, 1) \in C$, and $\mathbf{x} - \mathbf{p} = (2, 2)$, so $\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle = 2y_1 + 2(y_2 - 1) = 2(y_1 + y_2 - 1) \leq 0$ for $\mathbf{y} \in C$. By Theorem 17.3.1 (b) this is the nearest point, and $d(\mathbf{x}, C) = \|(2, 2)\| = 2\sqrt{2}$.

Exercise B2: Cutting off a point from a diamond.

Let $C = B_1 = \{\mathbf{y} \in \mathbb{R}^2 : |y_1| + |y_2| \leq 1\}$, with the dot product, and $\mathbf{x} = (2, 2)$.

- (a) Show that $P_C(\mathbf{x}) = (\frac{1}{2}, \frac{1}{2})$.
- (b) Hence find the strictly separating hyperplane of Theorem 17.3.2, and check it directly.
- (c) Find every supporting hyperplane of C at $(\frac{1}{2}, \frac{1}{2})$.

Solution.

- (a) Let $\mathbf{p} = (\frac{1}{2}, \frac{1}{2})$, which lies in C . Then $\mathbf{x} - \mathbf{p} = (\frac{3}{2}, \frac{3}{2})$ and, for $\mathbf{y} \in C$,

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{y} - \mathbf{p} \rangle = \frac{3}{2}(y_1 + y_2 - 1) \leq 0,$$

because $y_1 + y_2 \leq |y_1| + |y_2| \leq 1$. By Theorem 17.3.1 (b), $P_C(\mathbf{x}) = \mathbf{p}$.

- (b) With $\mathbf{a} = (\frac{3}{2}, \frac{3}{2})$, we have $\|\mathbf{a}\|^2 = \frac{9}{2}$ and $\langle \mathbf{a}, \mathbf{x} \rangle = 6$, so $c = 6 - \frac{9}{4} = \frac{15}{4}$. The hyperplane $\frac{3}{2}(y_1 + y_2) = \frac{15}{4}$ is the line $y_1 + y_2 = \frac{5}{2}$. Directly: on C , $y_1 + y_2 \leq 1 < \frac{5}{2}$, while $x_1 + x_2 = 4 > \frac{5}{2}$.
- (c) Let $\mathbf{a} \neq \mathbf{0}$ satisfy $\langle \mathbf{a}, \mathbf{y} \rangle \leq \langle \mathbf{a}, \mathbf{p} \rangle = \frac{1}{2}(a_1 + a_2)$ for all $\mathbf{y} \in C$. Taking $\mathbf{y} = \mathbf{e}_1$ and $\mathbf{y} = \mathbf{e}_2$ gives $a_1 \leq \frac{1}{2}(a_1 + a_2)$ and $a_2 \leq \frac{1}{2}(a_1 + a_2)$, that is, $a_1 \leq a_2$ and $a_2 \leq a_1$; so $a_1 = a_2$. Taking $\mathbf{y} = -\mathbf{e}_1$ gives $-a_1 \leq a_1$, so $a_1 \geq 0$, and $a_1 \neq 0$ since $\mathbf{a} \neq \mathbf{0}$. Conversely $\mathbf{a} = (a, a)$ with $a > 0$ works, by the inequality in (a). So the only supporting hyperplane at $\mathbf{p}$ is the line $y_1 + y_2 = 1$ through the edge containing $\mathbf{p}$.

Exercise B3: A unique extension.

On $\mathbb{R}^2$ with $\|\cdot\|_1$, let $U = \text{span}((1, 2))$ and $f(t(1, 2)) = 3t$. Show that f satisfies the hypothesis of Corollary 17.3.9, and find every extension F that the corollary allows.

Solution. $|f(t(1,2))| = 3|t| = |t| + 2|t| = \|t(1,2)\|_1$, so the hypothesis holds. Every $F \in (\mathbb{R}^2)^*$ is $F(\mathbf{x}) = c_1x_1 + c_2x_2$ (Theorem 4.1.1), and it extends f exactly when $F((1,2)) = 3$, that is, $c_1 + 2c_2 = 3$.

Next, $|F(\mathbf{x})| \leq \|\mathbf{x}\|_1$ for all $\mathbf{x}$ holds exactly when $|c_1| \leq 1$ and $|c_2| \leq 1$. Indeed, if both hold then $|c_1x_1 + c_2x_2| \leq |x_1| + |x_2|$; and if, say, $|c_1| > 1$, then $\mathbf{x} = \mathbf{e}_1$ gives $|F(\mathbf{e}_1)| > 1 = \|\mathbf{e}_1\|_1$.

So we need $c_1 = 3 - 2c_2$ with $|3 - 2c_2| \leq 1$ and $|c_2| \leq 1$. The first gives $1 \leq c_2 \leq 2$, the second $c_2 \leq 1$; hence $c_2 = 1$ and $c_1 = 1$. The only extension is $F(\mathbf{x}) = x_1 + x_2$.

17.3.8.3 C. Going deeper

Exercise C1: The projection does not stretch.

Let $C \subseteq V$ be non-empty, closed and convex. Prove that $\|P_C(\mathbf{x}) - P_C(\mathbf{y})\| \leq \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in V$.

Hint: write [Eq. 17.3.1](#) once at $\mathbf{x}$, tested with the point $P_C(\mathbf{y})$, and once at $\mathbf{y}$, tested with $P_C(\mathbf{x})$, and add.

Solution. Put $\mathbf{p} = P_C(\mathbf{x})$ and $\mathbf{q} = P_C(\mathbf{y})$. By [Theorem 17.3.1](#) (b), applied at $\mathbf{x}$ with the point $\mathbf{q} \in C$ and at $\mathbf{y}$ with the point $\mathbf{p} \in C$,

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{q} - \mathbf{p} \rangle \leq 0, \quad \langle \mathbf{y} - \mathbf{q}, \mathbf{p} - \mathbf{q} \rangle \leq 0.$$

The second is $\langle \mathbf{q} - \mathbf{y}, \mathbf{q} - \mathbf{p} \rangle \leq 0$. Adding, $\langle (\mathbf{x} - \mathbf{y}) + (\mathbf{q} - \mathbf{p}), \mathbf{q} - \mathbf{p} \rangle \leq 0$, that is,

$$\|\mathbf{q} - \mathbf{p}\|^2 \leq \langle \mathbf{y} - \mathbf{x}, \mathbf{q} - \mathbf{p} \rangle \leq \|\mathbf{y} - \mathbf{x}\| \|\mathbf{q} - \mathbf{p}\|$$

by Theorem 10.1.5. If $\mathbf{q} = \mathbf{p}$ there is nothing to prove; otherwise divide by $\|\mathbf{q} - \mathbf{p}\| > 0$. This proves $\|\mathbf{p} - \mathbf{q}\| \leq \|\mathbf{x} - \mathbf{y}\|$.

Exercise C2: Strict separation and distance.

Let $C_1, C_2 \subseteq V$ be non-empty, closed and convex, and put $\delta = \inf\{\|\mathbf{y} - \mathbf{x}\| : \mathbf{x} \in C_1, \mathbf{y} \in C_2\}$.

- (a) Prove that if C_1 and C_2 are strictly separated, then $\delta > 0$.
- (b) Prove that if $\delta > 0$, then C_1 and C_2 are strictly separated.
- (c) Hence deduce, without the computation of [Example \(A hyperbola and its asymptote\)](#), that the sets A and B there are not strictly separated.

Hint for (b): replace the difference set D of [Theorem 17.3.7](#) by its closure.

Solution.

- (a) Let $\mathbf{a} \neq \mathbf{0}$, c and $\varepsilon > 0$ strictly separate the sets, with C_1 on the side $\leq c - \varepsilon$. For $\mathbf{x} \in C_1$ and $\mathbf{y} \in C_2$, Theorem 10.1.5 gives

$$2\varepsilon \leq \langle \mathbf{a}, \mathbf{y} \rangle - \langle \mathbf{a}, \mathbf{x} \rangle = \langle \mathbf{a}, \mathbf{y} - \mathbf{x} \rangle \leq \|\mathbf{a}\| \|\mathbf{y} - \mathbf{x}\|,$$

so $\|\mathbf{y} - \mathbf{x}\| \geq 2\varepsilon/\|\mathbf{a}\|$. Hence $\delta \geq 2\varepsilon/\|\mathbf{a}\| > 0$.

- (b) Let $D = \{\mathbf{y} - \mathbf{x} : \mathbf{x} \in C_1, \mathbf{y} \in C_2\}$, which is convex by the computation in the proof of [Theorem 17.3.7](#). Every $\mathbf{d} \in D$ has $\|\mathbf{d}\| \geq \delta$, and so does every point of $\overline{D}$: if $\mathbf{d}_k \rightarrow \mathbf{d}$ with $\mathbf{d}_k \in D$, then $\|\mathbf{d}\| \geq \|\mathbf{d}_k\| - \|\mathbf{d}_k - \mathbf{d}\| \geq \delta - \|\mathbf{d}_k - \mathbf{d}\|$ for every k , and the right side tends to δ . So $\mathbf{0} \notin \overline{D}$, and $\overline{D}$ is non-empty, closed and convex by [Lemma 17.3.5](#) (a). By [Theorem 17.3.2](#) there is $\mathbf{b} \neq \mathbf{0}$ with $\langle \mathbf{b}, \mathbf{d} \rangle \leq -\|\mathbf{b}\|^2$ on $\overline{D} \supseteq D$. With $\mathbf{a} = -\mathbf{b}$, this is $\langle \mathbf{a}, \mathbf{x} \rangle + \|\mathbf{a}\|^2 \leq \langle \mathbf{a}, \mathbf{y} \rangle$ for all $\mathbf{x} \in C_1, \mathbf{y} \in C_2$, and the last paragraph of the proof of [Theorem 17.3.7](#) (a) turns it into strict separation with $\varepsilon = \frac{1}{2}\|\mathbf{a}\|^2$.
- (c) The points $(t, 1/t) \in A$ and $(t, 0) \in B$ are at distance $1/t$, so $\delta \leq 1/t$ for every $t > 0$, and $\delta = 0$. By (a), A and B are not strictly separated.